

Chapter 4 : Nutritional history taking and physical examination

4.1 Nutritional history taking

- Summary of all data regarding the patient's health from birth to present.
 - Direct vs. indirect

General Rules

1. Create an atmosphere of confidence and trust
 - a) Privacy
 - b) Comfortable environment
 - c) Eliminate haste/stress
2. Ask open questions
3. Let the patient choose his/her own words

History taking demands

- *Tact*
- *Patience*
- *Tolerance*
- *Sympathy and Understanding.*

Components of Hx taking

- *Identifying (Biographic)Data*
- *Source of the History,*
- *Chief Complaint(s),*
- *Present Illness,*
- *Past History,*
- *Family History,*
- *Personal and Social History,*
- *Review of Systems*

Component con'd ...

► **Identifying data**—such as

- age, gender, occupation, marital status
- *Source of the history—usually the patient, but can be family member or friend, letter of referral, or the medical record*
- If appropriate, establish *source of referral because a written report may be needed*

Component con'd ...

Reliability: ►

- Varies according to the patient's memory, trust, and mood

Chief Complaint(s)

- The one or more symptoms or concerns causing the patient to seek care
- These are the major symptoms for which the patient is seeking care or advice.
- They should be written using the words of the patient.
- The duration of the complaint should be specified

Component con'd ...

Present Illness

- Amplifies the *Chief Complaint*; describes how each symptom developed
- Includes patient's thoughts and feelings about the illness
- Pulls in relevant portions of the *Review of Systems*, called "pertinent positives and negatives
- May include *medications, allergies, habits of smoking and alcohol, which frequently are pertinent to the present illness*

Component con'd ...

- Present illness should be described as follows
- **Date of onset:**
- It is usually useful to start the history of the present illness with the phrase “the patient was perfectly or relatively well unit....”
- **Mode of onset, course and duration:**
 - ➡ Ask whether the onset was
 - abrupt or gradual
 - intermittent or persistent
 - short lived or constant, and
 - steady or increasing in severity.

Component con'd ...

Character and Location : If we take pain as an example,

- Stabbing
- Burning
- Pricking
- Aching

Component con'd ...

Exacerbating and Remitting Factors:

- what brings on and what makes the problem worse

The mode of arrival

- how the patient came to the health institution
- i.e., on a stretcher or walking.
- Also there should be mention of any color, strength, or weigh change.

Component con'd ...

Past History

- includes important illnesses from infancy onwards
- Lists childhood illnesses
- Lists adult illnesses with dates for at least four categories: medical, surgical, obstetric/gynecologic, and psychiatric
- Includes health maintenance practices such as immunizations, screening tests, lifestyle issues, and home safety

Component con'd ...

Family History

- Outlines or diagrams age and health, or age and cause of death, of siblings, parents, and grandparents
- Documents presence or absence of specific illnesses in family, such as hypertension, coronary artery disease, etc

Component con'd ...

Personal and Social History

- Describes educational level, family of origin, current household, personal interests, and lifestyle and daily function, ...

Review of Systems/*Functional Inquiry*

- Documents presence or absence of common symptoms related to each major body system
 - Skin, head, eyes, ears, nose, mouth, throat, respiratory tract, cardiovascular + lymphatics, GIT, urinary tract, genitalia, locomotor, nervous, psychological state, endocrine, allergies
 - Natural functions: voiding, defecation, eating habits/weight changes, sleep

Component con'd ...

should be recorded as follows

- ***General:*** *This includes history of recent weight change, weakness, fatigue, fever, etc*
- ***H.E.E.N.T. (Head, Eye, Ear, Nose, Mouth and Throat)***
- **Head:** Headache or injuries
- **Eyes:** double vision, blurring, photophobia, itching, pain, redness, excessive tearing, etc.

Component con'd ...

Ear: hearing problem, tinnitus, vertigo, earaches, discharge, etc

Nose: frequent colds, nasal stuffiness, nasal discharge or itching; nasal bleeding, etc.

- **Mouth and Throat: sore** tongue, frequent sore throat, and hoarseness of voice, dry mouth, oral thrush, dental carries, etc
- **Neck: pain, stiffness, swollen glands, "lumps", etc**

Component con'd ...

Lympho-glandular system

- includes enlarged glands, lumps in the breasts, and discharge from the nipples, goiter with or without heat or cold intolerance.
- descent of testis

Component con'd ...

Respiratory system

Includes

- Injury
- Cough
- Production of sputum (including the odor, color and amount)
- Hemoptysis
- Difficulty of breathing
- Wheezing
- Asthma
- Pulmonary tuberculosis
- Chest pain

Component con'd ...

- ***Cardiovascular system***
- Includes HX of Dyspnea (including degree of exercise tolerance)
- Palpitation
- Orthopnea (number of pillows required), paroxysmal nocturnal dyspnea
- Cough (dry or productive)
- Hemoptysis
- Chest pain (With character, location and radiation)
- Syncope(faintness)
- Hypertension
- Swelling of the feet

Component con'd ...

Gastro intestinal system: includes inquiry about history of

- Difficulty of swallowing, heartburn, nausea, vomiting, abdominal pain, constipation, diarrhea, food intolerance, excessive belching or passing of gas, frequency of bowel movement including the color and amount of stool passed, rectal bleeding, tarry stool, hemorrhoids, and jaundice

Component con'd ...

Genito – Urinary system:

- Urinary tract: History of flank pain, polyuria, nocturia, pain on micturation, passage of blood retention stream of urine, change in color of urine, urgency, frequency, hesitancy, dribbling, incontinence, or passage of stone during urination.

Component con'd ...

- Genitals:
- Males: history of hernias, discharge from the urethra or sores on the penis, testicular pain or masses, undescended testicles, past history of sexually transmitted diseases.
- Females: history of vaginal discharge and /or itching including their treatment.

Component con'd ...

- Menstrual history:
 - Age at menarche
 - Regularity
 - Frequency and duration of bleeding
 - Amount of bleeding
 - Last normal menstrual period
 - History of dysmenorrhea
 - Age at menopause
 - History of post-menopausal bleeding.
 - History of bleeding between periods or after intercourse

Component con'd ...

- ***Integumentary system***
- ***(Skin, Hair and Nails): History of dry or moist skin, rashes, ulcers, urticaria, hair distribution and pigmentary changes, changes in color and shape of the fingernails.***
- ***Musculo- skeletal system:*** History of bony deformities, joint pain and /or swelling, limping, loss of function of limbs or joints, leg swelling

Component con'd ...

- ***Central nervous system: History of fainting, seizures, weakness, paralysis,***
- numbness or loss of sensation, tingling sensation, tremor or other involuntary movements, insomnia, poor memory, headache, disturbance of speech, disturbance of sphincter control, delusion, hallucination, illusion, etc.

Note :- Be sure to distinguish *subjective* from *objective data*. Decide if your assessment will be comprehensive or focused

Subjective data

What the patient tells you
The history, from Chief Complaint
through Review of Systems

Objective data

What you detect during the examination
All physical examination findings

4.2:Physical examination

- **Approach**
- *is a systematic **sequence** of examination*
- recommended examining from the **patient's right side**
- Apply the techniques of inspection, palpation, auscultation, and percussion to each body region, but be sensitive to the whole patient.
- *Minimize the number of times to change position from supine to sitting, or standing to lying supine*

while examining the patient

- should take place with good lighting and in a quite environment.
- By words or gestures, be as clear as possible in your instructions.
- If possible try to demonstrate the patient what to do rather than giving verbal instructions alone
- Keep the patient informed as you proceed with your examination.
- While examining the patient, move “from head to toe.”

basic steps of physical examination

- Inspection
- Palpation
- Percussion
- Auscultation

components of comprehensive examination

- **General appearance:** Is the patient acutely sick, chronically sick looking or not sick looking at all? Is the patient in cardio respiratory distress or not?
- **Vital signs:**
 - Pulse (rate, volume, character, radio femoral delay)
 - Blood pressure (specify arm and the position it was taken)
 - Respiratory rate
 - Temperature (Specify the location it was taken)

Components

- H.E.E.N.T (head, eye, ear, nose, mouth and throat)
- Lymph glandular system
- Respiratory system
- Cardiovascular system
- Gastro intestinal system
- Genito urinary system
- Integumentary system
- Mesculo skeletal system
- Central nervous system.

HEENT

- Darken the room to promote pupillary dilation and visibility of the fundi

Head: Examine the hair, scalp, skull, and face.

Eyes: ***Check visual acuity and screen the visual fields. Note position*** and alignment of the eyes. Observe the eyelids. Inspect the sclera and conjunctiva of each eye. With oblique lighting, inspect each cornea, iris, and lens.

- Compare the pupils, and test their reactions to light.
- Assess extraocular movements. With an ophthalmoscope, inspect the ocular fundi.

HEENT...

- ***Ears: Inspect the auricles, canals, and drums. Check***
- auditory acuity. If acuity is diminished, check lateralization (Weber test) and compare air and bone conduction (Rinne test).
- ***Nose and sinuses: Examine the external nose; using a light and nasal speculum,***
- inspect nasal mucosa, septum, and turbinates. Palpate for tenderness of the frontal and maxillary sinuses

HEENT...

- ***Throat (or mouth and pharynx):***
- Inspect the lips, oral mucosa, gums, teeth, tongue, palate, tonsils, and pharynx. (*You may wish to assess the Cranial Nerves at this point in the examination.*)
- ***Neck. Move behind the sitting patient to feel the thyroid gland and*** to examine the back, posterior thorax, and lungs. Inspect and palpate the cervical lymph nodes. Note any masses or unusual pulsations in the neck.
- Feel for any deviation of the trachea. Observe sound and effort
- of the patient's breathing. Inspect and palpate the thyroid gland
- ***Back. Inspect and palpate the spine and muscles***

Chest

- ***Posterior Thorax and Lungs***
- Inspect and palpate the spine and muscles of the *upper back. Inspect, palpate, and percuss the chest.*
- Identify the level of diaphragmatic dullness on each side.
- Listen to the breath sounds; identify any adventitious (or added) sounds, and, if indicated, listen to transmitted voice sounds

Chest...

- ***Breasts, Axillae, and Epitrochlear Nodes. The patient is still sitting.***
- Move to the front again. ***In a woman, inspect the breasts with*** patient's arms relaxed, then elevated, and then with her hands pressed on her hips. ***In either sex, inspect the axillae and feel for the axillary*** nodes; feel for the epitrochlear nodes

Chest...

- ***Anterior Thorax and Lungs. The patient position is supine.***
- Ask the patient to lie down. Stand at the *right side of the patient's bed.*
- Inspect, palpate, and percuss the chest. Listen to the breath sounds, any adventitious sounds, and, if indicated, transmitted voice sounds

Chest...

- ***Cardiovascular System. Elevate head of bed to about***
- **30 degrees, adjusting as necessary to see the jugular venous pulsations.**
- Observe the jugular venous pulsations, and measure the jugular venous pressure in relation to the sternal angle.
- Inspect and palpate the carotid pulsations. Listen for carotid bruits

Abdomen

- ***Lower the head of the bed to the flat position.***
- ***The patient should be supine. Inspect, auscultate, and percuss. Palpate lightly, then deeply.***
- Assess the liver and spleen by percussion and then palpation.
- Try to feel the kidneys; palpate the aorta and its pulsations.
- If you suspect kidney infection, percuss posteriorly over the costovertebral angles

Peripheral Vascular System

- ***With the patient supine,***
- palpate the femoral pulses and, if indicated, popliteal pulses.
- Palpate the inguinal lymph nodes. Inspect for edema, discoloration, or ulcers in the lower extremities.
Palpate for pitting edema.
- **With the patient standing, inspect for varicose veins**

Lower Extremities

- ***Examine the legs, assessing the three***
- Systems, motor and sensory
- ***Nervous System. The patient is sitting or supine.***
- ***The examination*** of the nervous system can also be divided into the upper extremity examination (when the patient is still sitting) and the lower extremity examination (when the patient is supine) after examination of the peripheral nervous system.

Mental Status

- *If indicated and not done during the interview,*
assess orientation, mood, thought process,
thought content, abnormal perceptions, insight
and judgment, memory and attention,
information and vocabulary, calculating abilities,
abstract thinking, and constructional ability

Cranial Nerves

- *If not already examined, check sense of smell, funduscopic examination, strength of the temporal and masseter muscles, corneal reflexes, facial movements, gag reflex, strength of the trapezia and sternomastoid muscles, and protrusion of tongue*

Motor System

- *Muscle bulk, tone, and strength of major muscle groups.*
- *Cerebellar function: rapid alternating movements (RAMs),*
- point-to-point movements such as finger to nose (F → N) and heel
- to shin (H → S); gait. Observe patient's gait and ability to walk heel to toe, on toes, and on heels; to hop in place; and to do shallow knee bends
- Do a Romberg test; check for pronator drift

Sensory System

- *Pain, temperature, light touch, vibrations, and discrimination.*
- Compare right and left sides and distal with proximal areas on the limbs.

Additional Examinations

- ***The rectal and genital examinations are*** often performed at the end of the physical examination
- ***Male Genitalia and Hernias.*** *Examine the penis and scrotal contents.*
- Check for hernias
- ***Rectal Examination in Men.*** *The patient is **lying on his left side** for the rectal examination.*
- Inspect the sacrococcygeal and perianal areas.
- Palpate the anal canal, rectum, and prostate. (If the patient cannot stand, examine the genitalia before doing the rectal examination)

Additional Examinations...

- *Genital and Rectal Examination in Women. The patient is **supine in the lithotomy position**.*
- **Sit during the examination with** the speculum, then stand during bimanual examination of uterus, adnexa, and rectum.
- Examine the external genitalia, vagina, and cervix. Obtain a Pap smear.
- Palpate the uterus and adnexa. Do a bimanual and rectal examination.

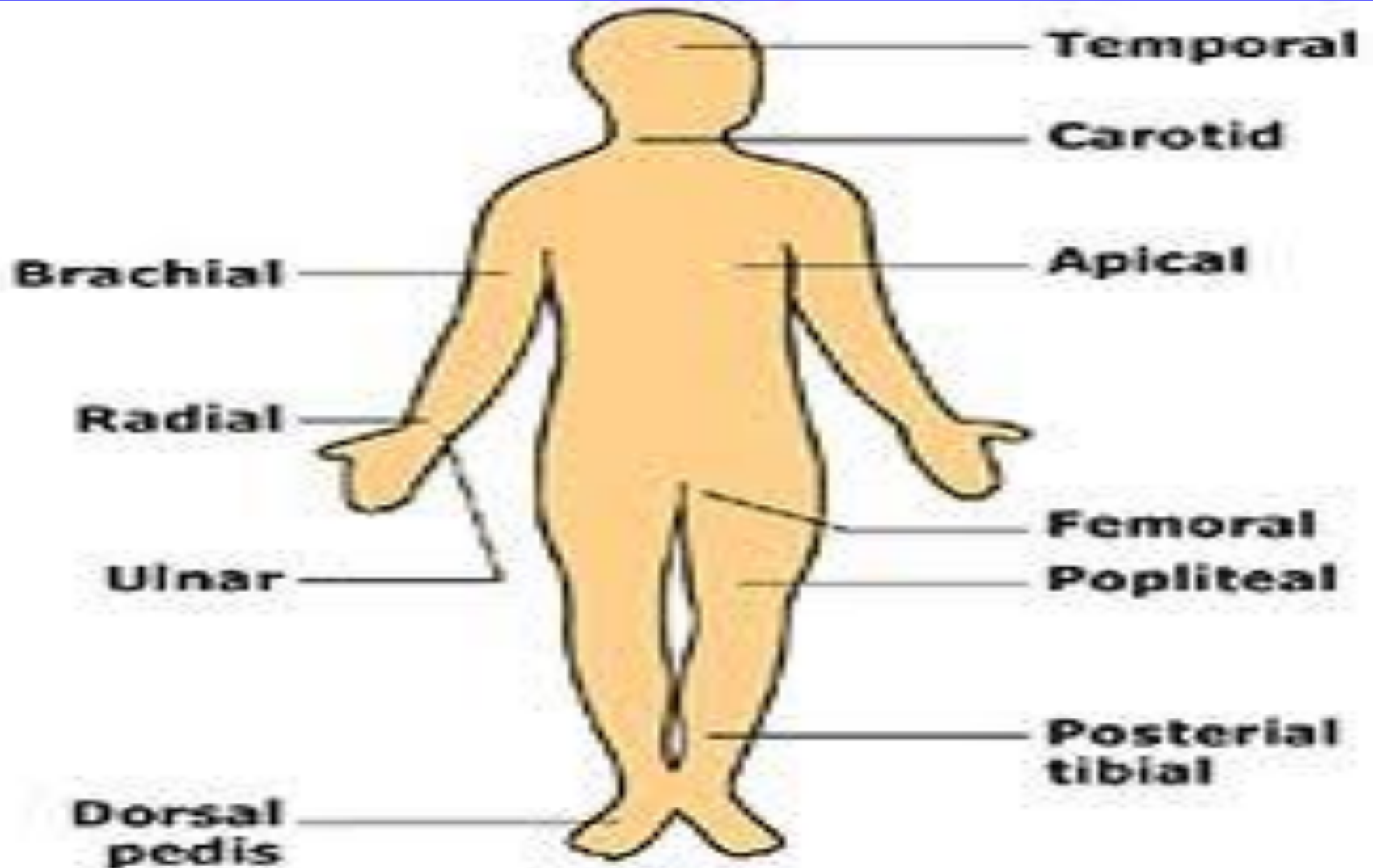
Chapter 5: Vital Sign

- Reflects essential body **functions**, including **heartbeat**, **breathing rate**, **temperature**, and **blood pressure**.
- You may watch, measure, or monitor vital signs to check level of **physical functioning**
- Vital signs are a group of the four to six most important medical signs that indicate the status of the body's **vital functions**.
- measurements help to assess general **physical** health of a person, give **clues** to possible **diseases**, and show **progress** toward **recovery**
- *It* can be measured in a medical setting, at home, at the site of a medical emergency, ...

Pulse rate

- pulse rate is a measurement of the **heart rate**, or the number of **times the heart beats** per minute.
- As the heart pushes blood through the arteries, the arteries expand and contract with the flow of the blood.
- Taking a pulse **not only** measures the **heart rate**, **but also** can indicate
 - Heart rhythm
 - Strength of the pulse

Pulse sites



How to check your pulse...

- Using the **first and second fingertips**, press firmly but gently on the arteries until you feel a pulse.
- Begin counting the pulse when the clock's second hand is on the 12.
- Count your pulse for **60 seconds** (or for 15 seconds and then multiply by four to calculate beats per minute).
- When counting, do not **watch the clock** continuously, but concentrate on the beats of the pulse.
- If **unsure** about your results, ask another person to count for you.



How to check your pulse...

- The normal pulse for healthy adults ranges from **60 to 100** beats per minute.
- The pulse rate may fluctuate and increase with **exercise, illness, injury, and emotions**.
- **Females** ages 12 and older, in general, tend to have faster heart rates than do **males**.
- Athletes, such as runners, who do a lot of cardiovascular conditioning, may have heart rates near **40 beats** per minute and experience no problems.

APICAL PULSE

- TAKEN WITH A STETHOSCOPE
- COUNTED BY PLACING THE STETHOSCOPE OVER THE HEART
- COUNTED FOR ONE FULL MINUTE
- THE HEART BEAT NORMALLY SOUNDS LIKE A *LUB-DUB*. EACH *LUB-DUB* IS COUNTED AS ONE HEARTBEAT.
- **DO NOT COUNT THE *LUB* AS ONE HEARTBEAT AND THE *DUB* AS ANOTHER.**
- THE APICAL PULSE IS TAKEN ON PATIENTS WHO HAVE HEART DISEASE , AN IRREGULAR PULSE RATE, OR TAKE MEDICATIONS THAT CAN AFFECT THE HEART.



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APICAL - RADIAL PULSE



THE APICAL AND RADIAL PULSE RATES SHOULD BE EQUAL

SOMETIMES THE HEART BEAT IS NOT STRONG ENOUGH TO CREATE A PULSE IN
THE RADIAL ARTERY

THIS WOULD CAUSE THE RADIAL PULSE TO BE LESS THAN THE APICAL PULSE
ONE PERSON COUNTS THE APICAL WHILE THE OTHER PERSON COUNTS THE
RADIAL

THE DIFFERENCE IN PULSES IS CALLED THE **PULSE DEFICIT**

NORMAL ADULT PULSE RATE IS – 60 TO 100 BEATS PER MIN.

TACHYCARDIA – HEART RATE OVER 100

BRADYCARDIA – HEART RATE BELOW 60

**REPORT ABNORMAL HEART RATES TO THE NURSE
IMMEDIATELY**

Table 19-2 Pulse Ranges for Different Ages	
Age	Pulse Rates per Minute
Birth to 4 weeks	80-180
4 weeks to 1 year	80-160
1 to 2 years	80-130
2 to 6 years	80-120
6 to 12 years	70-110
12 years and older	60-100

Respiratory rate

- Also called pulmonary **ventilation**, is the **act of breathing**.
- Is **the mechanism** the body uses to exchange gases between the **atmosphere & the blood & the cells**.
- Count each **inspiration & expiration** cycle as 1 respiration.
- Normal range is 12-20 breathes per min.
- **Inspiration** is the act of breathing **in**
- **Expiration** is the act of breathing **out**.
- Breathing is a **passive** process.

Respiration involves

- **Ventilation**: movement of gases **in** & **out of** the lungs
- **Diffusion**: movement of **carbon dioxide** & **oxygen** between the **alveoli** & the **red blood cells**.
- **Perfusion**: distribution of air to red blood cells & from the pulmonary **capillaries**

ABNORMAL RESPIRATIONS

TACHYPNEA – RESPIRATORY RATE OVER 20

BRADYPNEA – RESPIRATORY RATE BELOW 12

DYSPNEA – SHORTNESS OF BREATH – DIFFICULTY IN BREATHING

APNEA – NO BREATHING

HYPERVENTILATION – FAST AND DEEP RESPIRATIONS

HYPOVENTILATION – SLOW AND SHALLOW RESPIRATIONS

Blood pressure

- Blood pressure is the **force** of the blood pushing **against the artery walls** during **contraction** and **relaxation** of the heart.
- Each time the heart beats, it pumps blood into the arteries, resulting in the highest blood pressure as the heart **contracts**.
- When the heart **relaxes**, the blood pressure falls

Blood pressure...

- **Two** numbers are recorded when measuring blood pressure.
- The **higher** number, or **systolic** pressure, refers to the pressure inside the **artery** when the heart **contracts** and pumps blood through the body.
- The **lower** number, or **diastolic** pressure, refers to the pressure inside the artery when the heart is **at rest** and is **filling with blood**.
- Both the systolic and diastolic pressures are recorded as "mm Hg" (millimeters of mercury).

Blood pressure...

- This recording represents how high the mercury column in an old-fashioned manual blood pressure device (called a mercury manometer or sphygmomanometer) is raised by the pressure of the blood.
- Today, in the doctor's office is more likely to use a simple dial for this measurement

Blood pressure...

- High blood pressure, or hypertension, directly increases the risk of heart attack, heart failure, and stroke.
- With high blood pressure, the arteries may have an increased resistance against the flow of blood, causing the heart to pump harder to circulate the blood

Category of Blood pressure

- It is categorized as **normal**, **elevated**, or stage 1 or stage 2 high blood pressure:
- **Normal** blood pressure is systolic of less than 120 and diastolic of less than **80 (120/80)**
- **Elevated** blood pressure is systolic **of 120 to 129** and diastolic less than 80
- **Stage 1** high blood pressure is systolic is **130 to 139** or diastolic between **80 to 89**
- **Stage 2** high blood pressure is when systolic is **140** or higher or the diastolic is **90** or higher

Category of Blood pressure...

- These numbers should be used as a guide only.
- A **single blood** pressure measurement that is higher than normal is **not necessarily** an indication of a problem.
- The doctor will want to see **multiple blood** pressure measurements over **several days** or **weeks** before making a **diagnosis** of high blood pressure and **starting treatment**.

Normal range of blood pressure

Approximate Age Range	Systolic Range	Diastolic Range
1-4 years	80-110	50-80
3-5 years	80-110	50-80
6-13 years	85-120	55-80
13-18 years	95-140	60-90

Procedures of B/P measurement

- To measure , use a proper **sized blood pressure cuff**.
- The **length** of the cuff's bladder should be at least equal **to 80%** of the circumference of the **upper arm**.
- **Wrap** the cuff around the upper arm with the cuff's lower edge **one inch above** the **antecubital** fossa.
- Lightly press the **stethoscope's** bell over the brachial artery just below the cuff's edge.
- Some health care workers have **difficulty using the bell** in the antecubital fossa, so we suggest using the **bell** or the **diaphragm** to measure the blood pressure.
- Rapidly **inflate** the cuff to **180**mmHg.
- **Release air** from the cuff at a moderate rate (3mm/sec).

Procedures of B/P measurement...

- Listen with the stethoscope and simultaneously observe the **sphygmomanometer**.
- The first knocking sound (Korotkoff) is the subject's **systolic pressure**.
- When the knocking sound disappears, that is the **diastolic** pressure (such as 120/80).
- Record the pressure in **both arms** and **note** the difference; also record the **subject's position** (supine), which arm was used, and the cuff size (small, standard or large adult cuff).
- If the subject's pressure is elevated, **measure blood pressure two additional times**, waiting a few minutes between measurements.
- A BLOOD PRESSURE OF **180/120mmHg** OR MORE REQUIRES IMMEDIATE ATTENTION!

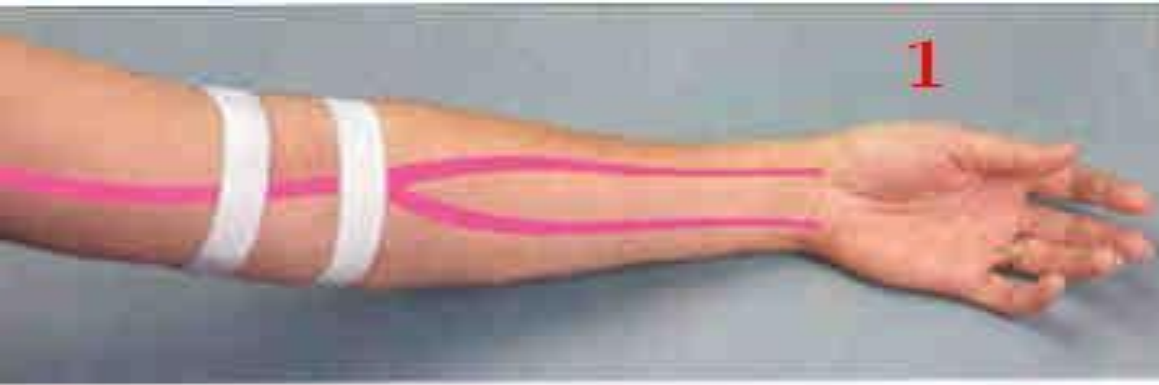
Sphygmomanometer



Pediatric



Adult



Precautions

- Aneroid and digital manometers may require periodic calibration.
- Use a **larger cuff** on obese or **heavily muscled** subjects.
- Use a **smaller** cuff for **pediatric** patients.
- For pediatric patients a lower blood pressure may indicate the presence of **hypertension**.
- Don't place the cuff over **clothing**.
- Flex and **support** the patient's arm.

Precautions...

- In some patients the Korotkoff sounds will disappear as the systolic pressure is bled down.
- After an interval, the Korotkoff sounds reappear.
- This interval is referred to as the "**auscultatory gap**."
- This pathophysiologic occurrence can lead to a marked under-estimation of systolic pressure if the cuff pressure is not elevated enough.
- It is for this reason that the rapid inflation of the blood pressure cuff to 180mmHg was recommended above.
- The "**auscultatory gap**" is felt to be associated with carotid atherosclerosis and a decrease in arterial compliance in patients with increased blood pressure.

Subject

- Position: supine, seated, standing.
- In seated position, the subject's arm should be flexed.
- The flexed elbow should be at the level of the heart.
- If the subject is **anxious**, wait a few minutes before taking the pressure.

Why should we monitor our blood pressure at home?

- For people with hypertension, home monitoring allows your doctor to monitor how much your blood pressure changes during the day, and from day to day.
- This may also help your doctor determine how effectively your blood pressure medication is working

What special equipment is needed to measure blood pressure?

- Either an aneroid monitor, which has a dial gauge and is read by looking at a pointer, or a digital monitor, in which the blood pressure reading flashes on a small screen, can be used to measure blood pressure

special equipment...

- Aneroid monitor is less expensive
- The cuff is inflated by hand by squeezing a rubber bulb.
- Some units even have a special feature to make it easier to put the cuff on with one hand.
- However, the unit can be easily damaged and become less accurate.
- Because the person using it must listen for heartbeats with the stethoscope

digital monitor

- The digital monitor is **automatic**, appearing on a **small screen**.
- recordings are easy to read, most **popular blood pressure** measuring device.
- It is also easier to use no need to listen to heartbeats through the stethoscope, good device for hearing-impaired.
- One disadvantage is that body movement or an **irregular heart rate** can change the accuracy.

Finger and wrist blood pressure monitors

- Tests have shown that finger and/or wrist blood pressure devices are not as accurate in measuring blood pressure as other types of monitors.
- In addition, they are more expensive than other monitors.

Before you measuring your blood pressure

American Heart Association guidelines for home blood pressure monitoring

- Don't smoke or drink coffee for 30 minutes before taking your blood pressure.
- Go to the bathroom before the test.
- Relax for 5 minutes before taking the measurement.
- Sit with your back supported (don't sit on a couch or soft chair).

Guidelines con; d

- Keep your feet on the floor uncrossed.
- Place your arm on a solid flat surface (like a table) with the upper part of the arm at heart level.
- Place the middle of the cuff directly above the bend of the elbow.
- Check the monitor's instruction manual for an illustration.

Guideline con;d

- Take multiple readings.
- When you measure, take 2 to 3 readings one minute apart and record all the results.
- Take your blood pressure at the same time every day, or as your healthcare provider recommends.
- Record the date, time, and blood pressure reading.
- Take the record with you to your next medical appointment.
- If your blood pressure monitor has a built-in memory, simply take the monitor with you to your next appointment.

Guideline con;d

- Call your provider if you have several high readings.
- Don't be frightened by a single high blood pressure reading, but if you get several high readings, check in with your healthcare provider.
- When blood pressure reaches a systolic (top number) of 180 or higher OR diastolic (bottom number) of 110 or higher, seek emergency medical treatment.

Body temperature

- The normal body temperature of a person varies depending on gender, recent activity, food and fluid consumption, time of day, and, in women, the stage of the menstrual cycle.
- Normal body temperature can range from 97.8 degrees F (or Fahrenheit, equivalent to 36.5 degrees C, or Celsius) to 99 degrees F (37.2 degrees C) for a healthy adult.

[Temperature]

- Two Types of Body Temperature
 - Core Temperature
 - Temperature of the deep tissues of the body
 - Remains relatively constant unless exposed to severe extremes in environmental temperature
 - Assessed by using a thermometer
 - Surface Temperature
 - Temperature of the skin
 - May vary a great deal in response to the environment
 - Assessed by touching the skin

ways of measurement

- **Orally.** Temperature can be taken by mouth using either the classic glass thermometer, or the more modern digital thermometers that use an electronic probe to measure body temperature.
- **Rectally.** Temperatures taken rectally (using a glass or digital thermometer) tend to be 0.5 to 0.7 degrees F higher than when taken by mouth.
- **Axillary.** Temperatures can be taken under the arm using a glass or digital thermometer.
- Temperatures taken by this route tend to be 0.3 to 0.4 degrees F lower than those temperatures taken by mouth.

ways of measurement...

- **By ear.** A special thermometer can quickly measure the temperature of the ear drum, which reflects the body's core temperature (the temperature of the internal organs).
- **By skin.** A special thermometer can quickly measure the temperature of the skin on the forehead.

ways of measurement...

WE NOW ALSO HAVE THE TEMPORAL SITE - FOREHEAD

MOST TEMPERATURES ARE TAKEN ORALLY

RECTAL TEMPERATURES ARE THE MOST ACCURATE

AXILLARY TEMPERATURES ARE THE LEAST ACCURATE

ways measurement...

- Body temperature may be abnormal due to fever (high temperature) or hypothermia (low temperature).
- A fever is indicated when body temperature rises about one degree or more over the normal temperature of 98.6 degrees Fahrenheit,
- Hypothermia is defined as a drop in body temperature below 95 degrees Fahrenheit

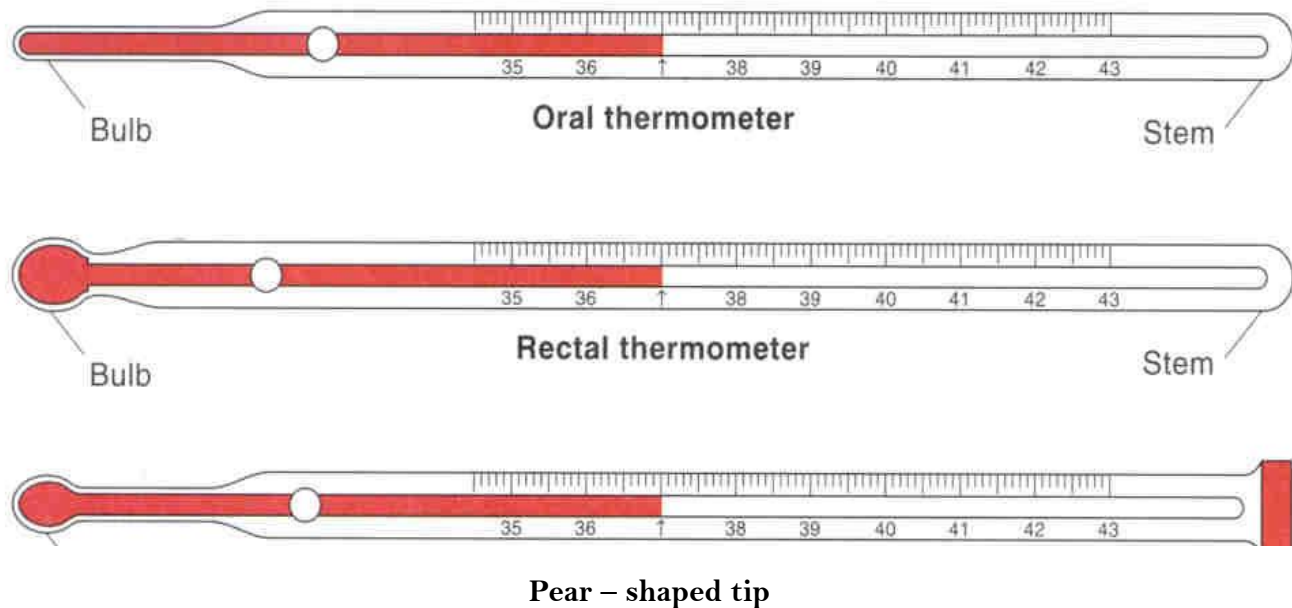
NORMAL BODY TEMPERATURE

SITE	NORMAL	RANGE
ORAL	98.6 °	97.6 ° TO 99.6 °
RECTAL	99.6 °	98.6 ° TO 100.6 °
AXILLARY	97.6 °	96.6 ° TO 98.6 °
TYMPANIC	98.6 °	98.6 °
TEMPORAL	98.6°	98.6°

TYPES OF THERMOMETERS

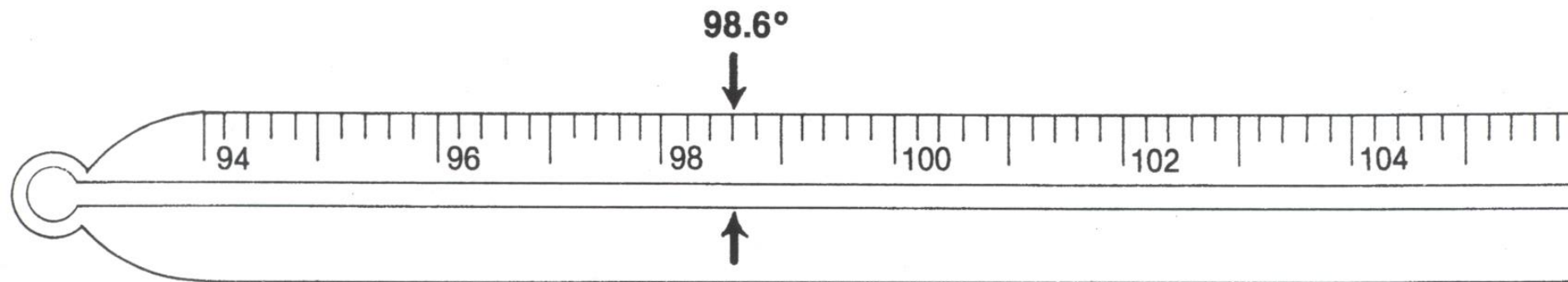
GLASS THERMOMETER

A SMALL HOLLOW GLASS TUBE THAT CONTAINS MERCURY OR A MERCURY-FREE SUBSTANCE IN A BULB AT ONE END. WHEN HEATED THE MERCURY RISES IN THE TUBE.



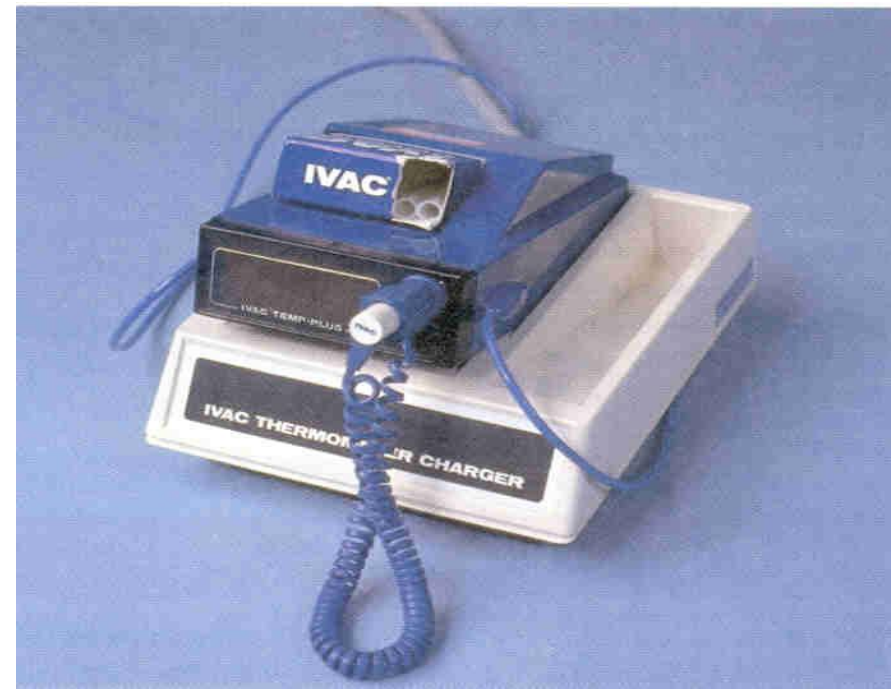
READING A GLASS THERMOMETER

- THE SCALE IS MARKED FROM 94° TO 108°
- THE LONG LINES REPRESENT ONE DEGREE
- THE SHORT LINES REPRESENT TWO TENTHS OF A DEGREE
- ONLY EVERY OTHER DEGREE IS MARKED WITH A NUMBER



ELECTRONIC THERMOMETER

- BATTERY OPERATED
- HAVE AN ORAL PROBE AND A RECTAL PROBE
- DISPOSABLE PROBE COVER IS PLACED ON THE PROBE
- THE TEMPERATURE REGISTERS IN ABOUT 30 SECONDS

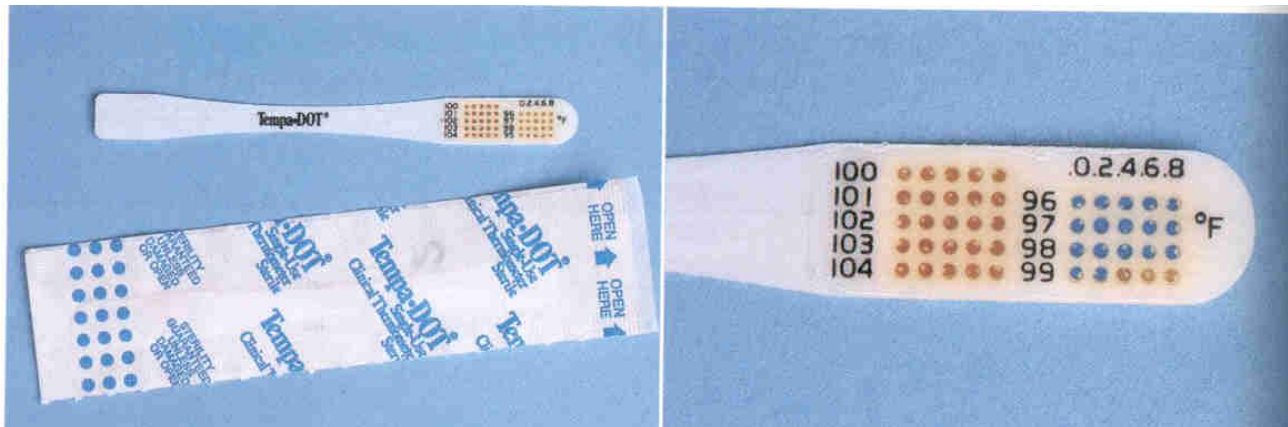


DIGITAL THERMOMETER

USE A DISPOSABLE SHEATH



DISPOSABLE ORAL THERMOMETER



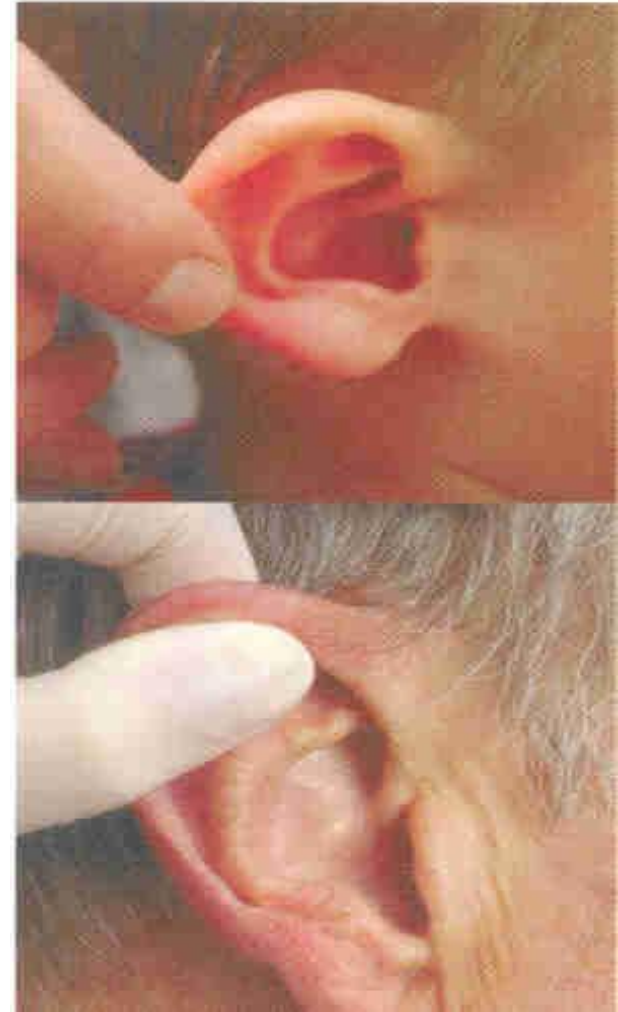
TYMPANIC THERMOMETER

- o MEASURES THE TEMPERATURE IN THE TYMPANIC MEMBRANE (EARDRUM)
- o FAST AND ACCURATE - 1 TO 3 SECONDS



INFANTS – PULL
THE EAR
STRAIGHT BACK

ADULTS AND
CHILDREN OVER
ONE YEAR –
PULL THE EAR UP
AND BACK



TAKING AN ORAL TEMPERATURE

GLASS THERMOMETER

- RINSE WITH COLD WATER
- CHECK THE THERMOMETER FOR BREAKS AND CHIPS
- SHAKE DOWN THE THERMOMETER SO THE MERCURY IS BELOW THE LINES AND NUMBERS
- PLACE A DISPOSABLE COVER ON THE THERMOMETER
- PLACE THE THERMOMETER UNDER THE PERSON'S TONGUE
- LEAVE THE THERMOMETER IN PLACE FOR 2 – 3 MINUTES
- IF THE PERSON HAS BEEN EATING, DRINKING, OR SMOKING, WAIT 15 MINUTES BEFORE TAKING TEMPERATURE



Fig. 19-3 The thermometer is held at the stem. It is read at eye level.

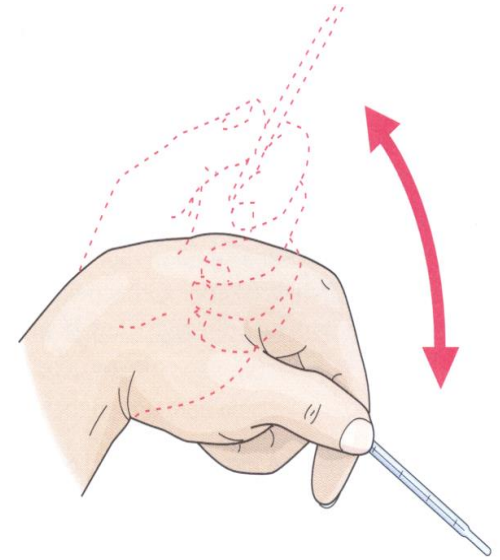


Fig. 19-4 The wrist is snapped to shake down the thermometer.

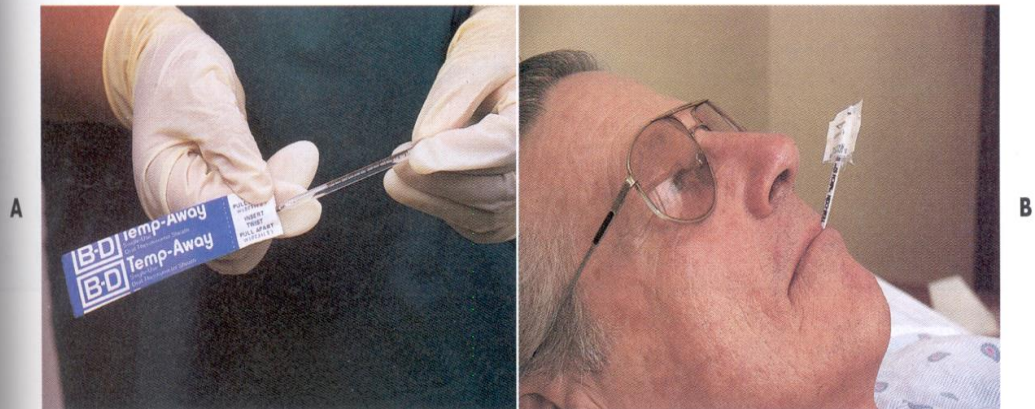


Fig. 19-5 A, The thermometer is inserted into a plastic cover. B, The person's temperature is taken with the thermometer in the plastic cover.

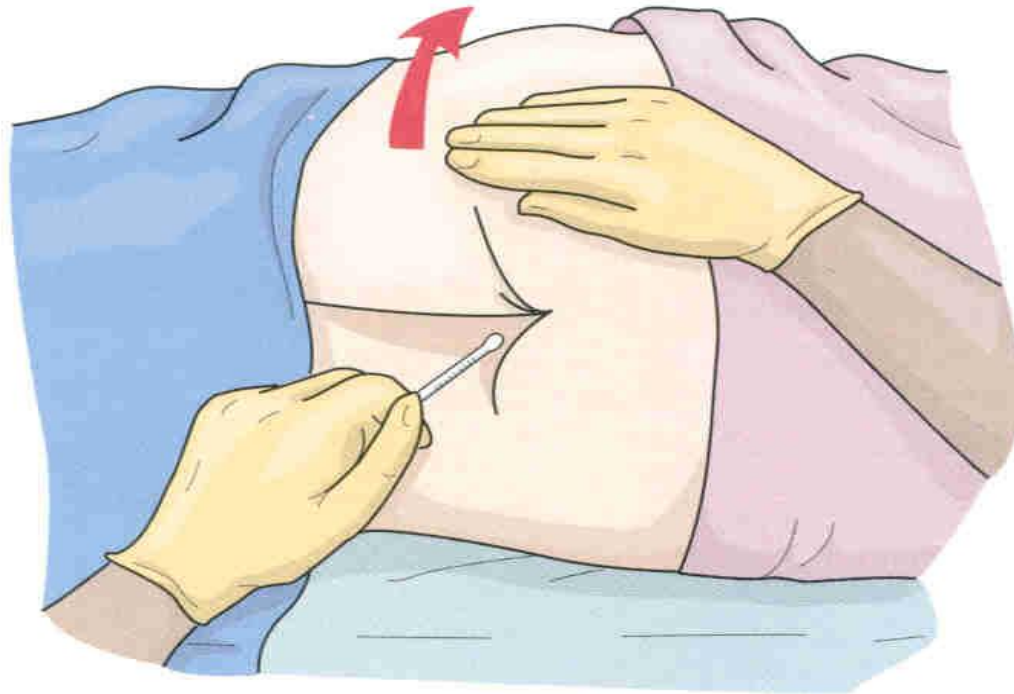
GUIDELINES FOR TAKING AN ORAL TEMPERATURE

DO NOT TAKE AN ORAL TEMPERATURE ON:

- o AN INFANT OR YOUNG CHILD (UNDER AGE 6)
- o AN UNCONSCIOUS PATIENT
- o A PATIENT THAT HAS HAD ORAL SURGERY OR AN INJURY TO THE FACE, NECK, NOSE, OR MOUTH
- o A PERSON RECEIVING OXYGEN
- o A PATIENT WITH A NASOGASTRIC TUBE IN PLACE
- o A PATIENT WHO IS CONFUSED OR RESTLESS
- o A PATIENT WHO IS PARALYZED ON ONE SIDE OF THE BODY
- o HAS A HISTORY OF SEIZURES
- o A PATIENT WHO BREATHES THROUGH THE MOUTH

TAKING A RECTAL TEMPERATURE

- o LUBRICATE THE THERMOMETER BEFORE INSERTING INTO THE RECTUM
- o PLACE THE PERSON IN A SIDE-LYING POSITION
- o INSERT THE THERMOMETER 1 INCH INTO THE RECTUM
- o HOLD THE THERMOMETER IN PLACE FOR 2 MINUTES
- o REMOVE THE DISPOSABLE COVER AND READ THE THERMOMETER

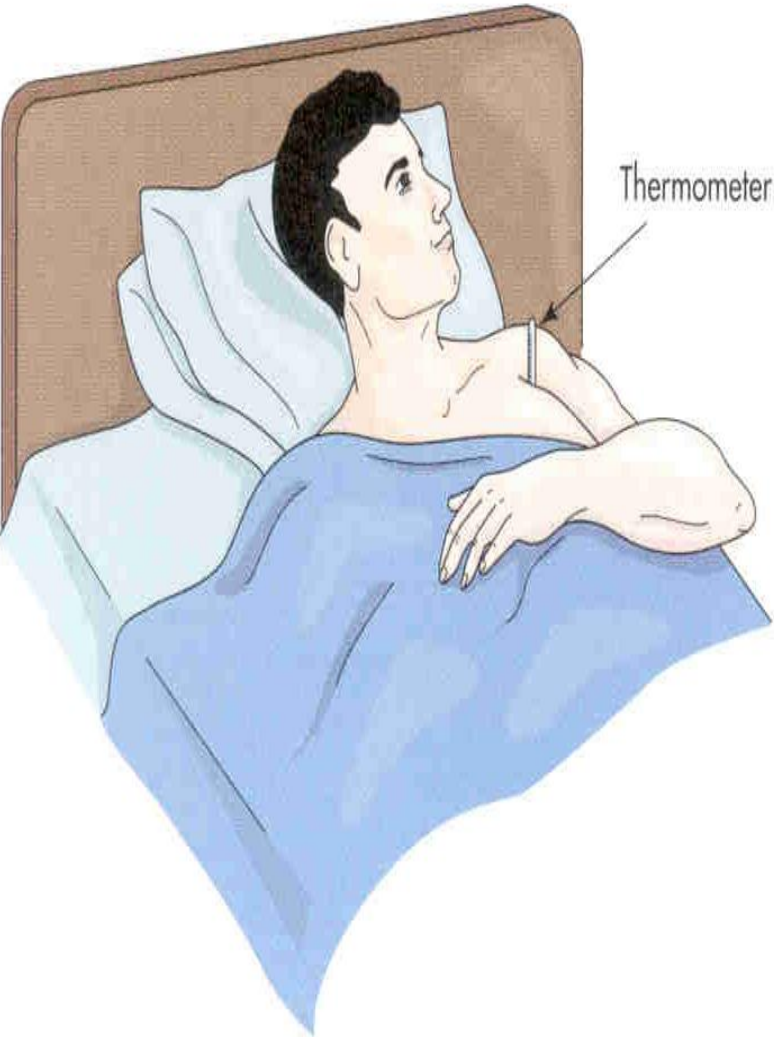


GUIDELINES FOR TAKING A RECTAL TEMPERATURE

DO NOT TAKE A RECTAL TEMPERATURE ON:

- o A PERSON WHO HAS HAD RECTAL SURGERY OR RECTAL INJURY
- o IF THE PERSON HAS DIARRHEA
- o IF THE PERSON IS CONFUSED OR AGITATED
- o IF THE PERSON HAS HEART DISEASE (STIMULATES THE VAGUS NERVE WHICH SLOWS THE HEART RATE)

TAKING AN AXILLARY TEMPERATURE



- TAKEN ONLY WHEN NO OTHER SITE CAN BE USED
- MAKE SURE THE UNDERARM IS CLEAN AND DRY
- THE ARM IS HELD CLOSE TO THE BODY
- YOU NEED TO HOLD THE THERMOMETER IN PLACE WHILE THE TEMPERATURE IS BEING TAKEN
- THE THERMOMETER IS LEFT IN PLACE FOR 10 MINUTES

Glass thermometers containing mercury

- Based on Environmental Protection Agency, mercury is a toxic substance that poses a threat to the health of humans, and the environment.
- So breaking, glass thermometers containing mercury should be removed from use and disposed of properly
- Contact your local health department, waste disposal authority, or fire department for information on how to properly dispose of mercury thermometers

Chapter 6: Legislation and standards in infection prevention and control

- Health care-associated infections (HAI) are one of the most common adverse events and major public health problem
- have a significant impact on morbidity, mortality
- However, they are preventable

Rules of universal precautions

- Consider ALL patients potentially infectious
- Assume ALL blood and body fluids and tissue to be potentially infectious
- Assume ALL unsterile needles and other sharps to be similarly contaminated

Standard Precautions

- should be followed in all patient care situations
- All staff should be informed of the need to report exposure to blood or potentially infectious body fluids

Common standard precautions

- Wash hands before and after all patient or specimen contact
- Handle the blood of **all patients** as potentially infectious
- Wear gloves for potential contact with blood and body fluids
- Prevent needle stick/sharp injuries
- Wear personal protective equipment (PPE) while handling blood or body fluids

Common standard precautions...

- Handle all linen soiled with blood and/or body secretion as potentially infectious
- Process all laboratory specimens as potentially infectious
- Wear a mask for TB and other contagious respiratory infections
- Maintain environmental cleanliness
- Follow proper waste disposal practices

Personal Protective Equipment(PPE) should be used by

- Healthcare workers who provide direct care to patients
- Support staff including medical aides, cleaners and laundry staff
- Laboratory staff, who handle patient specimens
- Family members who provide care to patients

Personal protective equipment

- Gloves
- Protective eye wear (goggles)
- Mask
- Apron
- Gown
- Boots or shoe covers
- Cap or hair cover

Proper Disposal of Needles and Sharps

- Needles and sharps are the commonest mode of transmission of blood-borne pathogens
- Precautions should be taken to prevent injuries by sharp instruments, especially hollow bore needles
- Needles should not be recapped, bent or broken by hand
- Do not overfill a sharps container

Good Practice for Safe Handling and Disposal of Sharps

- ALWAYS dispose of your own sharps
- NEVER **pass used** sharps directly from one person to another
- Protect fingers from injury by using forceps instead of fingers for guiding suturing

ISOLATION POLICIES AND PROCEDURES

- **Isolation** is used to separate ill person who have communicable disease from those who are healthy
- It restricts the movement of ill person to stop the spread of certain disease

Quarantine POLICIES AND PROCEDURES

- Is used to separate and restrict the movement of well person who may have been exposed to a communicable disease to see if they become ill

Removal of micro organism

- **Disinfection:** is a process removing most micro organisms from a defined object or surface, except **bacterial endospores**
- Classification based on their ability to destroy
 - If the agent is capable of rendering mycobacteria nonviable, it is termed as an intermediate level disinfectant
 - Agent which destroys only vegetative bacteria is termed a low level
 - It is safe to assume that all the other categories of microbes and efficacy **High level** disinfection

Sterilization

- Sterilization is defined as a process where **all microbes** are removed from a defined object, inclusive of bacterial endospores
- Methods of Sterilization Used
 - i. Steam autoclave
 - ii. Hot air oven

Time of sterilization

STERILIZATION	RECOMMENDATIONS
Hot Air Oven	160 °C for 1 hr, 180 °C for 30 min
Autoclave	Gravity-Displacement: • 30 min holding time at 121 °C • 1.1 kg/cm² or 15 lb/in² (PSI) Prevacuum : • 3 min holding time at 134 °C • 2.2 kg/cm² or 32 lb/in² (PSI)

Chapter 7: Cleaning and ventilation

- cleaning is necessary clean and free from dust and soil
- **Ninety percent** of microorganisms are present within “**visible dirt**”, and the purpose of routine cleaning is to eliminate this dirt
- frequency of cleaning and cleaning agents used for walls, floors, windows, beds,, and all reused medical devices must be specified
- Methods must be appropriate for potential contamination

Four hospital zones

- Zone A: No patient contact. Normal domestic cleaning is recommended (for example, administration, and library)
- Zone B: Care of patients who are not infected, and not highly susceptible, should be done by a procedure that does not raise dust

Four hospital zones...

- Zone C: Infected patients (isolation wards). Clean with a detergent or disinfectant solution, with separate cleaning equipment for each room
- Zone D: Highly susceptible patients (protective isolation) or protected areas such as operating suites, delivery rooms, intensive care units, premature baby units. Clean using a detergent or disinfectant solution and separate cleaning equipment

Cleaning Methods

Dry Methods

- ✓ Dust attractant mop
- ✓ Dry dusting is not recommended as it may disperse dust and micro-organisms
- ✓ Sweeping brushes must not be used in clinical areas as they disperse dust and micro-organisms

Wet Methods

- Scrubbing
- Mopping
- Damp dusting

General Principles to Follow when Cleaning

- Ensure any electrical equipment to be cleaned
- Refer to the health and safety data sheets before using any product
- Display warning signs in the area, ensuring all signs are visible
- Prepare a fresh solution of cleaning agent or disinfectant as required
- Wear suitable personal protective equipment
- Do not over-wet the floor

Ventilation

- Is the intentional introduction of outdoor air into a space
- is mainly used to control indoor air quality by diluting and displacing indoor pollutants;
- it can also be used to control indoor temperature, humidity, and air motion to benefit thermal comfort,.

Methods of ventilation

- mechanical ventilation,
- natural ventilation
- mixed-mode ventilation (hybrid ventilation).

Mechanical ventilation

- Is the intentional fan driven flow of outdoor air into a building.
- This systems may include supply fans (which push outdoor air into a building), exhaust fans (which draw air out of building and thereby cause equal ventilation flow into a building), or a combination of both.
- Mechanical ventilation is often provided by equipment that is also used to heat and cool a space.

Natural

- is the intentional passive flow of outdoor air into a building through **planned** openings (such as doors, and windows).
- Natural ventilation does not require mechanical systems to move outdoor air.
- Instead, it relies entirely on passive physical phenomena, such as wind pressure, or the stack effect.

Mixed-mode ventilation

- systems use both mechanical and natural processes.
- The mechanical and natural components may be used at the same time, or at different times of day, or in different seasons of the year.
- Ventilation is typically described as separate from infiltration

Chapter 8:Selected and common clinical procedures

Cardiopulmonary Resuscitation (CPR)

- consists of the use of chest compressions and artificial ventilation to maintain circulatory flow and oxygenation during cardiac arrest

Cardiopulmonary resuscitation (CPR)...

- is a lifesaving technique useful in many emergencies, including a heart attack or near drowning, in which someone's breathing or heartbeat has stopped
- CPR can keep oxygenated blood flowing to the brain and other vital organs until more definitive medical treatment can restore a normal heart rhythm

Before starting CPR

- Is the environment safe for the person?
- Is the person conscious or unconscious?
- If the person appears unconscious, tap or shake his or her shoulder and ask loudly, "Are you OK?"
- If the person doesn't respond and two people are available, have one person call 0000 or the local emergency number and get the AED, if one is available, and have the other person begin CPR
- If you are alone and have immediate access to a telephone, call 0000 or your local emergency number before beginning CPR.

Different types resuscitation



Chest compressions



Open the airway



Rescue breathing

Procedures

- Note the overlapping hands placed on the **center of the sternum**, with the rescuer's arms extended.
- Chest compressions are to be delivered at a rate of at least 100 compressions per minute.



Compressions: Restore blood circulation

1. Put the person on his or her back on a firm surface.
2. Kneel next to the person's neck and shoulders.
3. Place the heel of one hand over the center of the person's chest, between the nipples. Place your other hand on top of the first hand. Keep your elbows straight and position your shoulders directly above your hands.

Compressions: Restore blood circulation...

4. Use your upper body weight (not just your arms) as you push straight down on (compress) the chest at least 2 inches (approximately 5 centimeters) but not greater than 2.4 inches (approximately 6 centimeters). Push hard at a rate of 100 to 120 compressions a minute.

- If you haven't been trained in CPR, continue chest compressions until there are signs of movement or until emergency medical personnel take over. If you have been trained in CPR, go on to opening the airway and rescue breathing.

Airway: Open the airway

- If you're trained in CPR and you've performed 30 chest compressions, open the person's airway using the head-tilt, chin-lift maneuver.
- Put your palm on the person's forehead and gently tilt the head back.
- Then with the other hand, gently lift the chin forward to open the airway

Breathing: Breathe for the person

- Rescue breathing can be mouth-to-mouth breathing or mouth-to-nose breathing (Not recommended)
 - With the airway open (using the head-tilt, chin-lift maneuver), pinch the nostrils shut for mouth-to-mouth breathing and cover the person's mouth with yours, making a seal.
1. Prepare to give two rescue breaths. Give the first rescue breath — lasting one second — and watch to see if the chest rises. If it does rise, give the second breath. If the chest doesn't rise, repeat the head-tilt, chin-lift maneuver and then give the second breath. Thirty chest compressions followed by two rescue breaths is considered one cycle.

Breathing...

3. Resume chest compressions to restore circulation.

4. As soon as an automated external defibrillator (AED) is available, apply it and follow the prompts. Administer one shock, then resume CPR — starting with chest compressions — for two more minutes before administering a second shock.

Continue CPR until there are signs of movement or emergency medical personnel take over.

NG tube insertion and removal

- is usually considered as the first line in artificial nutrition support for patients with a functioning gastrointestinal tract whose nutritional needs cannot be met by diet alone or by diet and nutritional supplements

Aims of carrying out this procedure

- To provide adequate nutrition
- To maintain patient safety
- To ensure comfort and co-operation of the patient
- To monitor patients for complications of naso-gastric feeding
- To administer feed as prescribed by the Dietitian

Who needs an NG

- Surgical clients
- Ventilated client
- Neuromuscular impairment .
- Clients who are unable to maintain adequate oral intake to meet metabolic demands.

Measurement



IV secure

- **Intravenous therapy** is treatment that infuses intravenous solutions, medications, blood, or blood products directly into a vein
- Intravenous therapy is an effective and fast-acting way to administer fluid or medication treatment in an emergency situation, and for patients who are unable to take medications orally. Approximately 80% of all patients in the hospital setting will receive intravenous therapy

Purpose

- To replace fluids and electrolytes
- To administer medications, including chemotherapy, anesthetics, and diagnostic reagents:
- To administer blood or blood products:
- The To deliver nutrients and nutritional supplements:

Shock position

- A sudden and severe setback due to any reason
- circulatory shock or 'cardiovascular collapse' is characterized by a reduction in circulatory blood volume and results in inadequate tissue perfusion.
- It is caused by severe hemorrhage, extensive trauma or burns, large myocardial infarction, massive pulmonary embolism, and microbial sepsis

Classification

1. Hypovolemic shock: absolute reduction in blood volume due to

(a) Loss of blood

(b) Loss of water and electrolytes

(c) Loss of plasma

Classification...

2. Cardiogenic shock: results from myocardial pump failure

3 Septic shock:is caused by systemic microbial infection

4. Neurogenic shock:results without any loss of blood volume

5. Anaphylactic shock:hypersensitivity reactions can lead to release of histamine or other substances, (produce vasodilation increase in capillary permeability)

Signs and symptoms

(1) Increased thirst

2) Body temperature

- In hypovolemic and cardiogenic shock:
Decreased body temperature
- In septic shock: Increased body temperature

(3) Increased HR

(4) Decreased urine output

(5) Restlessness and apprehension

treatment in shock

- Replacement therapy
 - Blood and plasma transfusion
 - Dextran solution
- Treatment with sympathomimetic drugs

Other therapies

- Treatment by the head-down position
- Oxygen therapy
- Treatment with glucocorticoids